The Serial Position Effect in Free Recall:

A *psychtools* Usage Demonstration

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Author Note

All data in this document are simulated for illustrative purposes.

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Abstract

We report a simulated free-recall experiment in which 30 participants studied 12-item word lists and recalled them immediately afterwards. Recall proportions showed the classic U-shaped serial position curve, with significant primacy and recency effects. This document illustrates the use of the *psychtools* LaTeX package; all data are invented.

Keywords: serial position effect, free recall, psychtools, LaTeX

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Introduction

The serial position effect (SPE) is one of the most robust phenomena in memory research: items at the beginning and end of a study list are recalled better than items in the middle (e.g., Murdock, 1962). The primacy advantage is typically attributed to extra rehearsal of early items, whereas the recency advantage reflects retrieval from a short-term or working-memory buffer. Atkinson and Shiffrin's (1968) modal model formalised this distinction into separate short-term and long-term stores.[1]

Evidence for separate mechanisms comes from the dissociation between primacy and recency under filled-delay conditions (e.g., Glanzer & Cunitz, 1966): a distractor task inserted between study and test selectively eliminates the recency effect while leaving primacy intact.

Raw data and non-parametric replications are provided in the S (see Tables S1 and the alternative analyses in Section SM2).

Method

Note: All data in this document are simulated. Any resemblance to real experimental results is coincidental.

Participants

Thirty hypothetical participants (M age = 22.4 years, SD = 3.1, range 18–34) were recruited from a psychology subject pool. Two participants were excluded for recall proportions more than 2.5 standard deviations below the group mean.[2]

[1] (Footnote will be removed) An alternative account invokes temporal distinctiveness — add citation — rather than dual stores.

[2] (Footnote will be removed) Exclusion criterion should be verified against preregistration document.

Materials

Stimuli were 12 common English nouns (mean log frequency = 2.34, $SD = 0.41$) drawn from the MRC Psycholinguistic Database.

Procedure

Words were presented on screen at a rate of one per second. Immediately after the last word, participants typed their free recall responses for 60 seconds. Order of recall was unconstrained.

Results

A one-way repeated-measures ANOVA on recall proportion across the 12 serial positions revealed a significant main effect of position, $F(11, 319) = 8.43$, $p < .001$, $\eta_p^2 = .23$. Overall, $M = .47$ of words were recalled ($SD = .17$, $SE = .03$).

Planned contrasts confirmed a primacy effect (positions 1–3 vs. 4–9): $t(29) = 4.21$, $p < .001$, Cohen's $d = 0.77$, $CI_{.95} = [0.38, 1.15]$. A recency effect (positions 10–12 vs. 4–9) was also significant: $t(29) = 5.63$, $p < .001$, Cohen's $d = 1.03$, $CI_{.95} = [0.57, 1.47]$.

The serial position curve is shown in Figure 1. A two-panel summary of the primacy and recency regions is displayed in Figure 2.

(Header will be removed) Effect of Presentation Rate

This section will discuss how varying the inter-stimulus interval modulates the magnitude of the primacy effect. Data collection for a follow-up experiment is ongoing.

Discussion

The simulated data reproduce the classic U-shaped serial position curve, consistent with Atkinson and Shiffrin's (1968) dual-store account. Both primacy and recency were large (Cohen's $d > 0.75$), in line with typical effect sizes reported in the literature (e.g., Murdock, 1962).

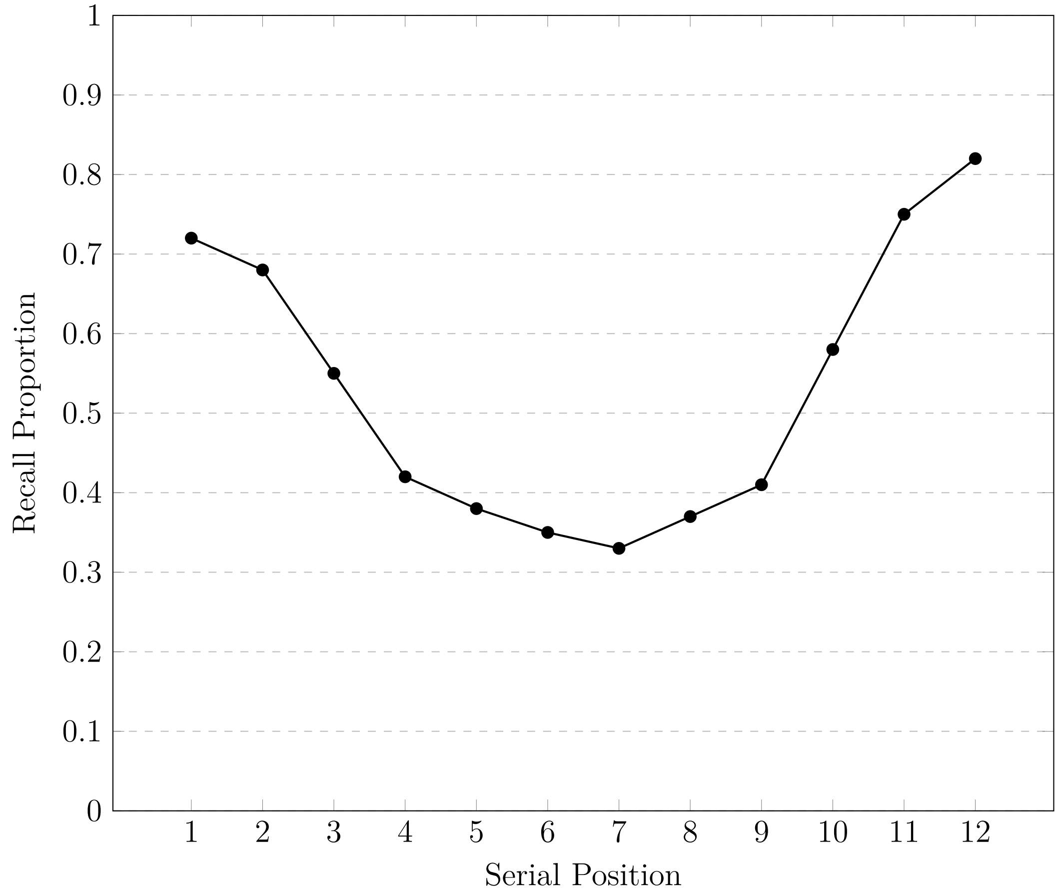


Figure 1. Simulated serial position curve ($N = 30$, error bars omitted). *Note.* Data are invented for illustrative purposes.

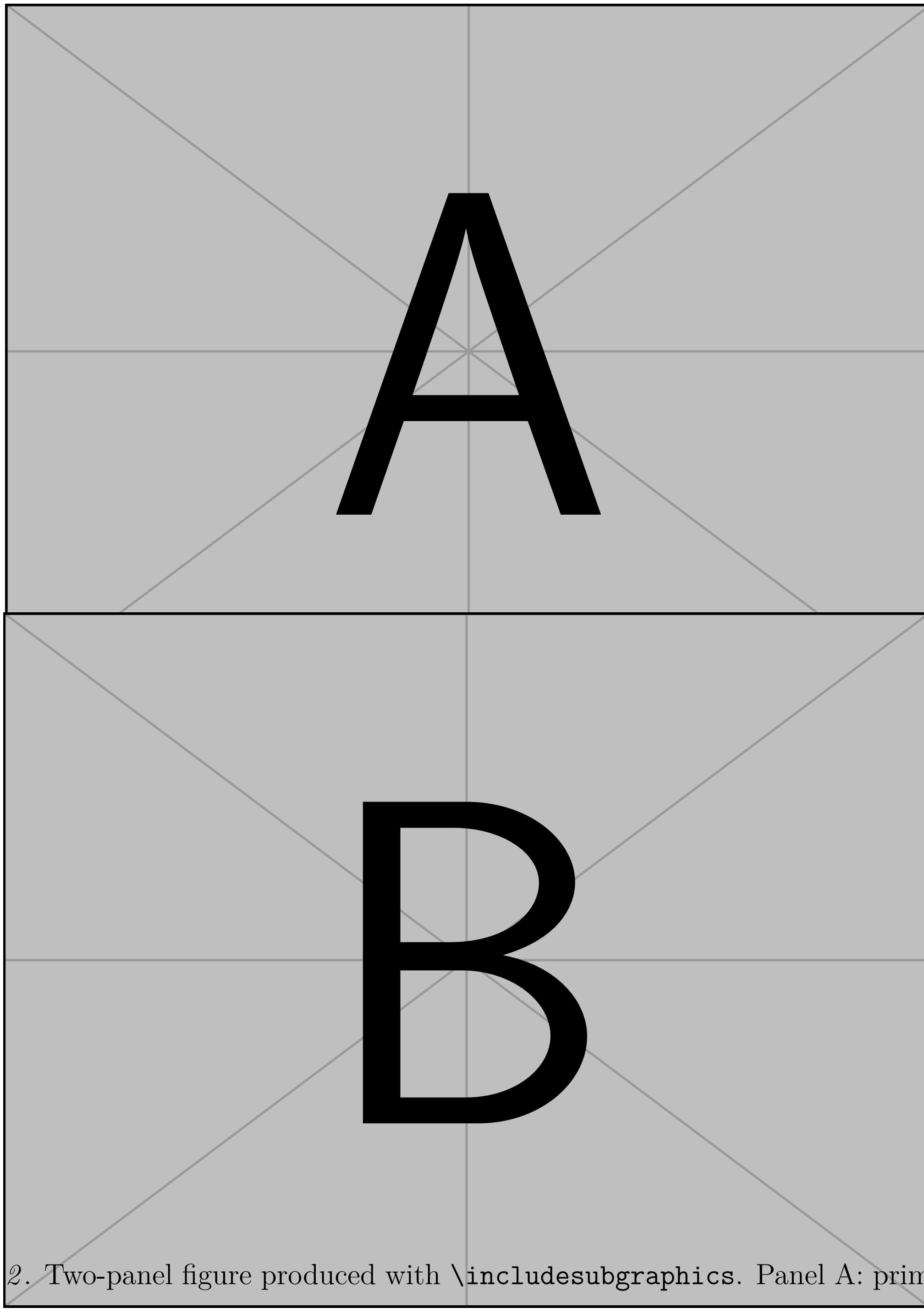
We agree with Reviewer 2 that the implications for working-memory capacity limits deserve further discussion, and have expanded this paragraph accordingly.

(Header will be removed) Comparison With Delayed Recall

Glanzer and Cunitz's (1966) findings predict that a filled delay should selectively reduce recency. A direct comparison of immediate and delayed conditions is planned for future work.

(Header will be removed) Potential Confounds. Word frequency and imageability should be matched across serial positions in future studies to rule out item-level confounds. Per Reviewer 1: add analysis of frequency effects here.

A



B

Figure 2. Two-panel figure produced with `\includesubgraphics`. Panel A: primacy region (positions 1–3). Panel B: recency region (positions 10–12). *Note*. Placeholder images shown; replace with real figures.

References

Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. *The Psychology of Learning and Motivation*, *2*, 89–195.

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S1 Raw Data

Table S1 presents the simulated mean recall proportions by serial position.

Table S1

Simulated Mean Recall Proportions by Serial Position

Position	*M*	*SD*
1	.72	.17
2	.68	.18
3	.55	.16
4	.42	.15
5	.38	.14
6	.35	.14
7	.33	.15
8	.37	.15
9	.41	.16
10	.58	.17
11	.75	.16
12	.82	.15

Note. All data are simulated for illustrative purposes.

S2 Alternative Analyses

S2.1 Non-Parametric Tests

S2.1.1 (Header will be removed) Confirm wording with statistician. As a robustness check, we replicated the primacy and recency contrasts using non-parametric tests. The Wilcoxon signed-rank test confirmed the primacy effect: $W(30) = 412$, $Z = 3.42$, $p < .001$. The Mann–Whitney test confirmed the recency advantage: $U = 156$, $Z = 4.11$, $p < .001$. The standard error of the position-1 recall proportion was $SE = .031$. For completeness, the effect-size estimates were $\eta^2 = .21$ (eta-squared) and $\eta_p^2 = .23$ (partial eta-squared) from the parametric ANOVA reported in the main text.